



Laboratory Report # 4

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Date Completed: October 4, 2024

Laboratory Exercise Title: Dataflow Modeling of Combinational Circuits

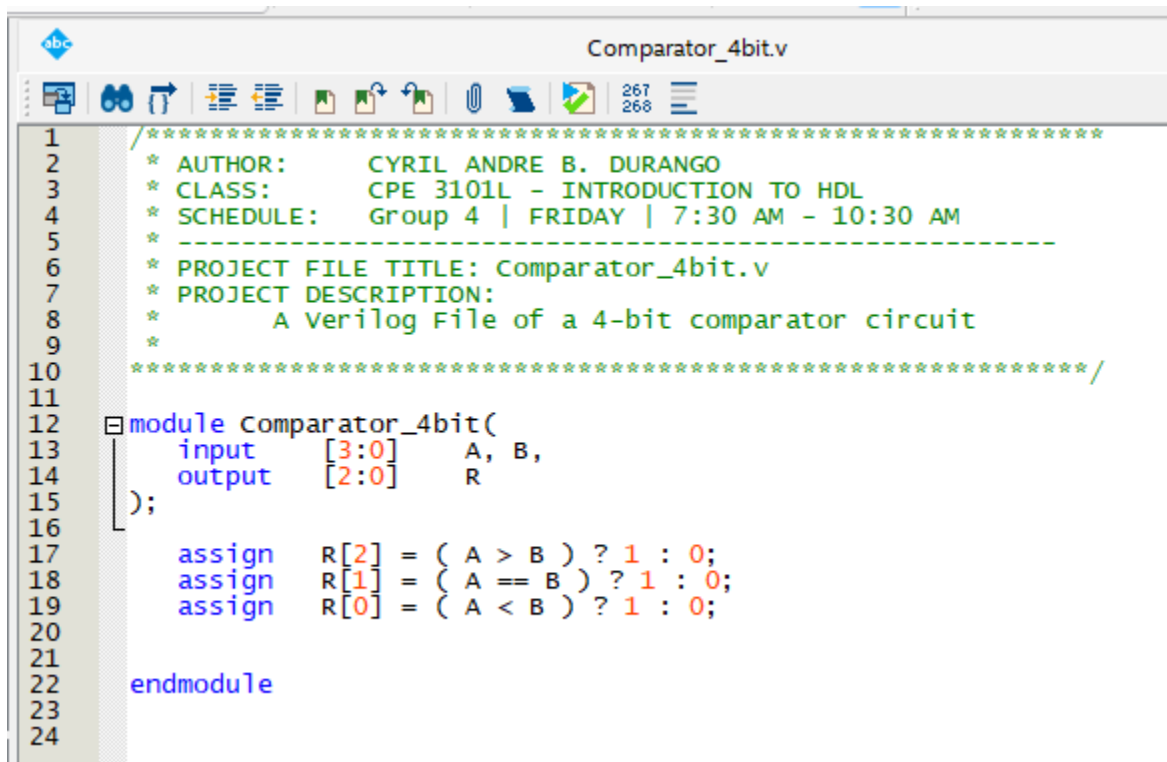
Target Course Outcomes:

CO1: Create descriptions of digital hardware components such as in combinational and sequential circuits using synthesizable Verilog HDL constructs.

CO2: Verify the functionality of HDL-based components through design verification tools.

Exercise 4A: 4-Bit Comparator. Under a new project, create a Verilog HDL description of a 4-bit comparator using dataflow modeling. Refer to the entity diagram and output details below for reference. Synthesize the design and simulate to verify the results. Show both a simulation log (standard output) and simulation waveform (timing diagram) in the laboratory report.

The 4-bit Comparator Verilog description file utilizes logical & arithmetic operators and ternary operators rather than building the circuit ground up with gate primitives.



```
1  /*****
2  * AUTHOR:      CYRIL ANDRE B. DURANGO
3  * CLASS:       CPE 3101L - INTRODUCTION TO HDL
4  * SCHEDULE:    Group 4 | FRIDAY | 7:30 AM - 10:30 AM
5  * -----
6  * PROJECT FILE TITLE: Comparator_4bit.v
7  * PROJECT DESCRIPTION:
8  *   A Verilog File of a 4-bit comparator circuit
9  *
10 /*****/
11
12 module Comparator_4bit(
13     input  [3:0]  A, B,
14     output [2:0]  R
15 );
16
17     assign R[2] = ( A > B ) ? 1 : 0;
18     assign R[1] = ( A == B ) ? 1 : 0;
19     assign R[0] = ( A < B ) ? 1 : 0;
20
21
22 endmodule
23
24
```

Figure 1. Verilog File of a 4-bit Comparator Circuit



```
tb_Comparator_4bit.v
1
2
3  * AUTHOR:      CYRIL ANDRE B. DURANGO
4  * CLASS:       CPE 3101L - INTRODUCTION TO HDL
5  * SCHEDULE:    Group 4 | FRIDAY | 7:30 AM - 10:30 AM
6  *
7  * -----
8  * PROJECT FILE TITLE: tb_Decoder_3by8.v
9  * PROJECT DESCRIPTION:
10 *   A Testbench Verilog File of a
11 *   3x8 Decoder Circuit
12 *
13 *****
14
15 `timescale 1 ns/1 ps
16 module tb_Comparator_4bit();
17
18     reg [3:0] A,B;
19     wire [2:0] R;
20
21     Comparator_4bit UUT(A,B,R);
22
23     initial begin
24
25         A = 4'd15; B = 4'd14; #10
26         A = 4'd11; B = 4'd11; #10
27         A = 4'd13; B = 4'd9; #10
28         A = 4'd10; B = 4'd3; #10
29         A = 4'd0; B = 4'd4; #10
30         A = 4'd1; B = 4'd12; #10
31         A = 4'd4; B = 4'd0; #10
32         A = 4'd7; B = 4'd7; #10
33         A = 4'd6; B = 4'd8; #10
34         A = 4'd5; B = 4'd1; #10
35
36     $stop;
37     end
38 endmodule
```

Figure 2. Testbench File of a 4-bit Comparator Circuit

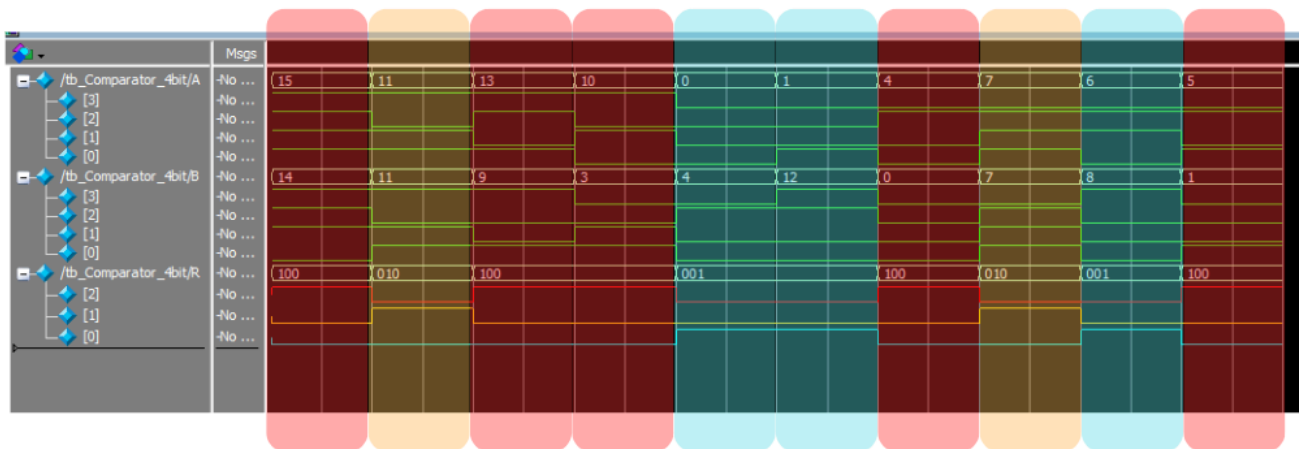
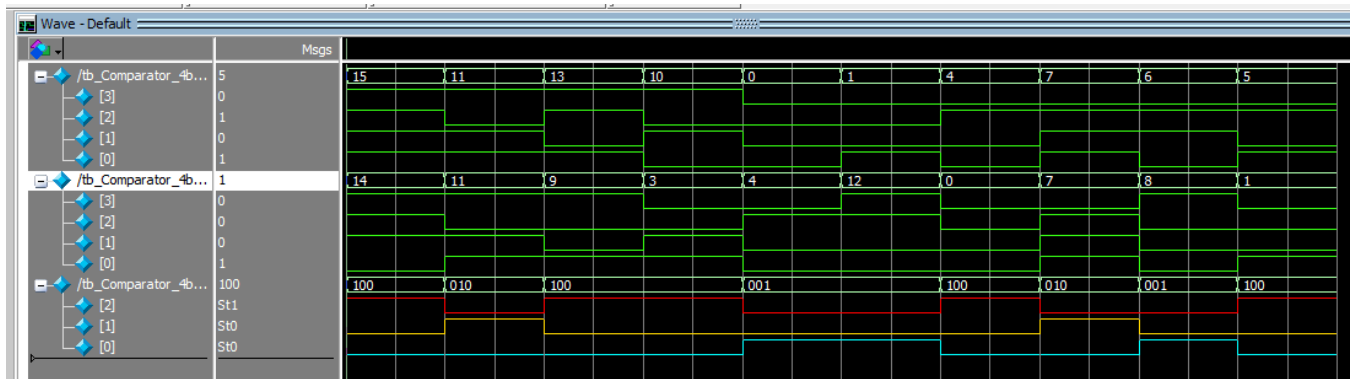
The screenshot displays the Quartus Prime Lite Edition interface. The main window shows the 'Flow Summary' tab, which provides a detailed overview of the compilation process. The summary indicates that the compilation was successful, with the following key statistics:

Category	Value
Flow Status	Successful - Fri Oct 11 07:36:43 2024
Quartus Prime Version	20.1.1 Build 720 11/11/2020 SJ Lite Edition
Revision Name	Comparator_4bit
Top-level Entity Name	Comparator_4bit
Family	MAX 10
Device	10M02DCU324A6G
Timing Models	Final
Total logic elements	9
Total registers	0
Total pins	11
Total virtual pins	0
Total memory bits	0
Embedded Multiplier 9-bit elements	0
Total PLLs	0
UFM blocks	0
ADC blocks	0

The 'Messages' window at the bottom shows the following messages:

- 12021 Found 1 design units, including 1 entities, in source file tb_comparator_4bit.v
- 12127 elaborating entity "comparator_4bit" for the top level hierarchy
- 10230 Verilog HDL assignment warning at comparator_4bit.v(17): truncated value with size 32 to match size of target (1)
- 10230 Verilog HDL assignment warning at comparator_4bit.v(18): truncated value with size 32 to match size of target (1)
- 10230 Verilog HDL assignment warning at comparator_4bit.v(19): truncated value with size 32 to match size of target (1)
- 286030 Timing-Driven Synthesis is running
- 16010 Generating hard_block partition "hard_block:auto_generated_inst"
- 21057 Implemented 20 device resources after synthesis - the final resource count might be different
- Quartus Prime Analysis & Synthesis was successful. 0 errors, 4 warnings

Figure 3. Verified Summary Compilation for the Verilog and Testbench Files of a 4-bit Comparator



red - $A > B$
yellow - $A = B$
cyan - $A < B$

Figure 4. Generated Waveform of the 4-bit Comparator Testbench File, with and without annotations.

The generated waveform demonstrates the logic behind a comparator. In the output waveform, red denotes that A is greater than B, yellow indicates that A is equal to B, and cyan denotes the A is less than B. As seen in the highlighted version, each of the highlighted columns corresponds to a logic-high output in the output waveforms. Those highlighted in red can be seen that A is greater than B, with its corresponding output of 100 that indicating so. The same logic can be applied to the rest of the highlighted areas.



Exercise 4B: n-Bit 4-to-1 Line Multiplexer. Design a dataflow description of an n-bit 4x1 multiplexer. Use a parameter to change the value of n (use n = 4 as the default value). Refer to the entity diagram and truth table below for reference. Synthesize the design.

```
1  /******************************************************************
2  * AUTHOR:      CYRIL ANDRE B. DURANGO
3  * CLASS:      CPE 3101L - INTRODUCTION TO HDL
4  * SCHEDULE:   Group 4 | FRIDAY | 7:30 AM - 10:30 AM
5  *
6  * PROJECT FILE TITLE: NBit_4by1_Multiplexer.v
7  * PROJECT DESCRIPTION:
8  *   A Verilog File of a N-Bit 4-by-1
9  *   Multiplexer circuit
10 *
11 *
12 *
13 *
14 *
15 *
16 *
17 *
18 *
19 *
20 *
21 *
22 *
23 *
24 *
25 *
26 *
27 *
28 *
29 *
30 *
31 */
32
33 module NBit_4by1_Multiplexer
34 #(
35     parameter N = 4
36 )
37 (
38     input  [N-1:0] A, B, C, D,
39     input  [1:0] S,
40     output [N-1:0] Y
41 );
42
43     assign Y = (S[0] == 0 && S[1] == 0) ? A :
44               (S[0] == 1 && S[1] == 0) ? B :
45               (S[0] == 0 && S[1] == 1) ? C :
46               D;
47
48 endmodule
```

Figure 5. Verilog File of an N-bit 4-by-1 Multiplexer

Quartus Prime Lite Edition - Z:/HDL Program Files/Exercise4B/NBit_4by1_Multiplexer - NBit_4by1_Multiplexer

File Edit View Project Assignments Processing Tools Window Help

Project Navigator Hierarchy Entity Instance Table of Contents Flow Summary

Entity Instance: MAX 10: 10M02DCU32446G NBit_4by1_Multiplexer 0 (8)

Table of Contents: Flow Summary, Flow Settings, Flow Non-Default Global Settings, Flow Elapsed Time, Flow OS Summary, Flow Log, Analysis & Synthesis, Flow Messages, Flow Suppressed Messages

Tasks: Compilation, Task, Compile Design, Analysis & Synthesis, Fitter (Place & Route), Assembler (Generate program), Timing Analysis

Flow Summary: Flow Status: Successful - Sat Oct 05 10:02:36 2024; Quartus Prime Version: 20.1.1 Build 720 11/11/2020 5J Lite Edition; Revision Name: NBit_4by1_Multiplexer; Top-level Entity Name: NBit_4by1_Multiplexer; Family: MAX 10; Device: 10M02DCU32446G; Timing Models: Final; Total logic elements: 8; Total registers: 0; Total pins: 22; Total virtual pins: 0; Total memory bits: 0; Embedded Multiplier 9-bit elements: 0; Total PLAs: 0; UFM blocks: 0; ADC blocks: 0

Messages: Running Quartus Prime Analysis & Synthesis; Command: quartus_map --read_settings_files-on --write_settings_files-off NBit_4by1_Multiplexer -c NBit_4by1_Multiplexer; 18236 Number of processors has not been specified which may cause overloading on shared machines. Set the global assignment NUM_PARALLEL_PROCESSORS in your QSF to an appropriate value for best performance; 20030 Parallel compilation is enabled and will use 10 of the 10 processors detected; 12021 Found 1 design units, including 1 entities, in source file nbit_4by1multiplexer.v; 12021 Found 1 design units, including 1 entities, in source file tb_4bit_4by1multiplexer.v; 12127 Elaborating entity "NBit_4by1_Multiplexer" for the top level hierarchy; 286030 Timing-Driven Synthesis is running; 16010 Generating hard_block partition "hard_block:auto_generated_inst"; 21057 Implemented 30 device resources after synthesis - the final resource count might be different; Quartus Prime Analysis & Synthesis was successful. 0 errors, 1 warning

Figure 6. Verified Summary Compilation for the N-bit 4-by-1 Multiplexer



```
tb_4Bit_4by1_Multiplexer.v  tb_8Bit_4by1_Multiplexer.v
1  /*****
2  * AUTHOR:      CYRIL ANDRE B. DURANGO
3  * CLASS:       CPE 3101L - INTRODUCTION TO HDL
4  * SCHEDULE:    Group 4 | FRIDAY | 7:30 AM - 10:30 AM
5  *
6  * PROJECT FILE TITLE: tb_4Bit_4by1_Multiplexer.v
7  * PROJECT DESCRIPTION:
8  *   A Testbench Verilog File of a
9  *   4-Bit 4X1 Multiplexer
10 *
11 *****/
12
13 `timescale 1 ns/1 ps
14 module tb_4Bit_4by1_Multiplexer();
15
16     reg    [3:0] A,B,C,D;
17     reg    [1:0] S;
18     wire   [3:0] Y;
19
20     NBit_4by1_Multiplexer #(N(4))
21     UUT(A,B,C,D,S,Y);
22
23     initial begin
24
25         A = 4'd15 ; B = 4'd10 ; C = 4'd5 ; D = 4'd0 ; S = 2'd0; #10
26         S = 2'd1; #10
27         S = 2'd2; #10
28         S = 2'd3; #10
29
30         $stop;
31     end
32
33 endmodule
34
```

Figure 7. Testbench File of a 4-bit 4-by-1 Multiplexer Circuit

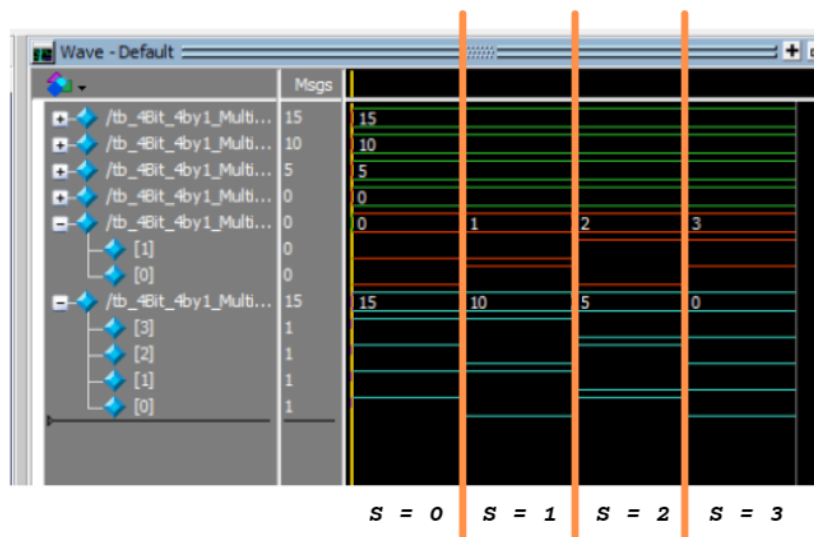


Figure 8. Annotated Generated Waveform of a 4-bit 4-by-1 Multiplexer Circuit

The above waveform shows the selector characteristics of the multiplexer. When S = 0, the output waveform generated is the binary equivalent of what is inputted in the 0th location. The same logic applies to the rest of the inputs and its corresponding selector.



```
1  /*****  
2  * AUTHOR:      CYRIL ANDRE B. DURANGO  
3  * CLASS:       CPE 3101L - INTRODUCTION TO HDL  
4  * SCHEDULE:    Group 4 | FRIDAY | 7:30 AM - 10:30 AM  
5  *  
6  * PROJECT FILE TITLE: tb_8Bit_4by1_Multiplexer.v  
7  * PROJECT DESCRIPTION:  
8  *   A Testbench Verilog File of a  
9  *   8-Bit 4x1 Multiplexer  
10 *  
11 *****/  
12  
13 `timescale 1 ns/1 ps  
14 module tb_8Bit_4by1_Multiplexer();  
15  
16     reg    [7:0] A,B,C,D;  
17     reg    [1:0] S;  
18     wire   [7:0] Y;  
19  
20     NBit_4by1_Multiplexer #(N(8))  
21         UUT(A,B,C,D,S,Y);  
22  
23     initial begin  
24  
25         A = 8'd255 ; B = 8'd128 ; C = 8'd200 ; D = 8'd171 ; S = 2'd0; #10  
26         S = 2'd1; #10  
27         S = 2'd2; #10  
28         S = 2'd3; #10  
29  
30         $stop;  
31     end  
32  
33 endmodule  
34
```

Figure 9. Testbench File of an 8-bit 4-by-1 Multiplexer Circuit

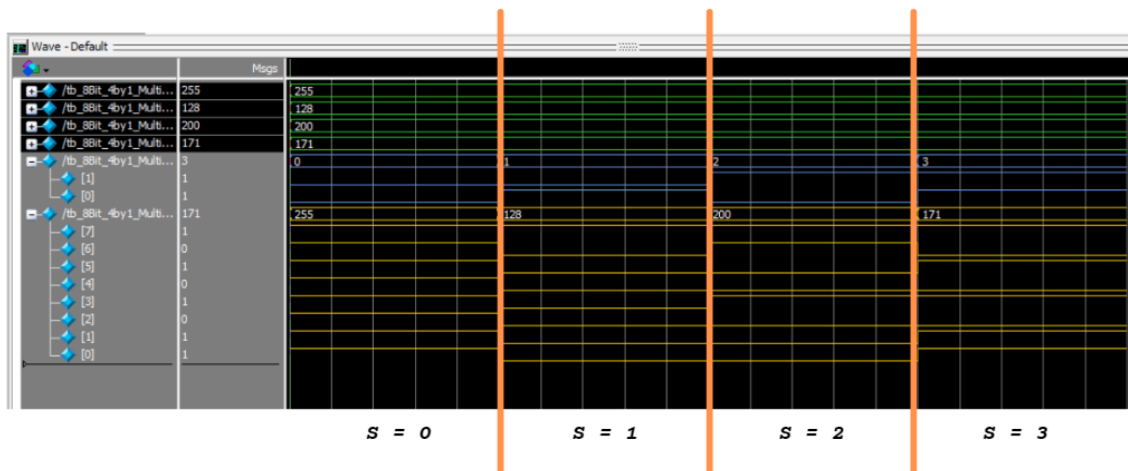


Figure 10. Annotated Generated Waveform of a 4-bit 4-by-1 Multiplexer Circuit

The above waveform shows the selector characteristics of the multiplexer. When S = 0, the output waveform generated is the binary equivalent of what is inputted in the 0th location. The same logic applies to the rest of the inputs and its corresponding selector.

It can also be observed in both generated waveforms for the 4-bit and 8-bit versions of the 4-by-1 Multiplexer Circuit, the input groups are always triggered. It will only be outputted by the time the selector is configured to its corresponding input group. A is outputted when S = 0, B is outputted when S = 1, C is outputted when S = 2, and D is outputted when S = 3